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Course: GEOG 420 - Interactive and Animated Cartography
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Spatial - temporal Analysis of Kidney Disease Mortality in Illinois (2008-2022)

Introduction

Kidney disease is a major public health issue in the United States, contributing significantly to morbidity and mortality. Kidney disease, also called chronic kidney disease (CKD) is a long-term condition characterized by the slow and progressive decline in kidney function. Healthy kidneys filter waste and excess fluids from the blood, regulate blood pressure, and maintain essential electrolytes. When damaged, these functions weaken, causing waste to accumulate in the body. CKD often develops due to diabetes, hypertension, infections, or other underlying health conditions, and can eventually lead to kidney failure or increased mortality risk. Understanding the spatial and temporal patterns of kidney-related deaths provides valuable insight into regional health disparities and supports targeted health interventions.

This project analyzes kidney disease mortality rates across Illinois counties from 2008 to 2022 using GIS-based static maps, interactive web maps, and animated cartography techniques. The primary goal was to visualize how mortality rates change over time and space and highlight counties with persistently high or increasing rates.

Animated choropleth mapping was used to help reveal spatial–temporal dynamics, while an interactive web map allowed users to explore individual county-level mortality trends.

Data Sets and Data Sources

This project utilized statewide county boundary data and county-level kidney disease mortality rates with corresponding population counts.

Dataset	Description	Source
Illinois county boundary shapefile	Spatial units for mapping	USA shapefile package provided via Blackboard
Kidney disease mortality data (2008–2022)	Annual deaths by county	Illinois Department of Public Health (IDPH) Statistical Access
Population counts	Annual county population estimates	IDPH/Processing from original dataset
Projection	NAD 1983 StatePlane Illinois East (US Feet)	Applied in ArcGIS Pro

Dataset	Description	Source
Software	ArcGIS Pro, ArcGIS Online, PowerPoint	Used for analysis, publishing, and animation

The mortality rate was calculated per 100,000 population to normalize data and enable fair comparison between counties of varying sizes.

Methodology

The project followed a guided cartographic workflow. First, the Illinois county shapefile was added to ArcGIS Pro. Mortality and population data were prepared in a CSV file and joined to the county layer using the county name attribute. A new field was computed to derive mortality rates per 100,000 residents.

A choropleth map was created for 2008 using a five-class Natural Breaks (Jenks) classification method. These class break values were recorded and applied manually to all other years (2009–2022) to ensure consistency and accurate temporal comparison. Map elements such as legend, title, and north arrow were fixed in position to avoid frame movement during animation.

Individual county rate maps for each year were exported as JPEGs and sequenced in Microsoft PowerPoint to produce a time-series animation in MP4 format. Additionally, the joined dataset was published to ArcGIS Online, where pop-ups were configured to display county names and mortality rates for 2008-2020, creating an interactive experience for end-users.

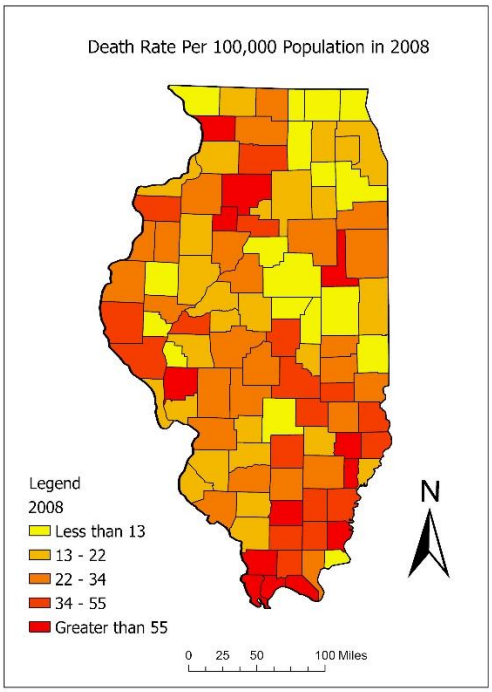
Results

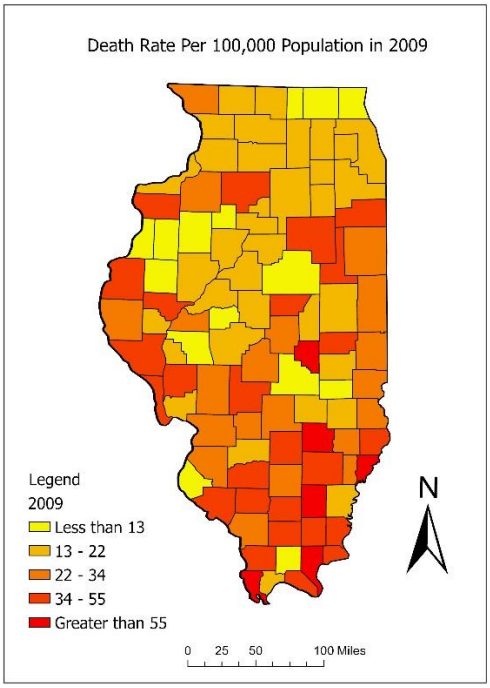
This project produced three primary outputs:
(1) static maps showing annual county mortality rates,
(2) a chronological map animation (2008-2022), and
(3) an interactive ArcGIS Online map for detailed county exploration.

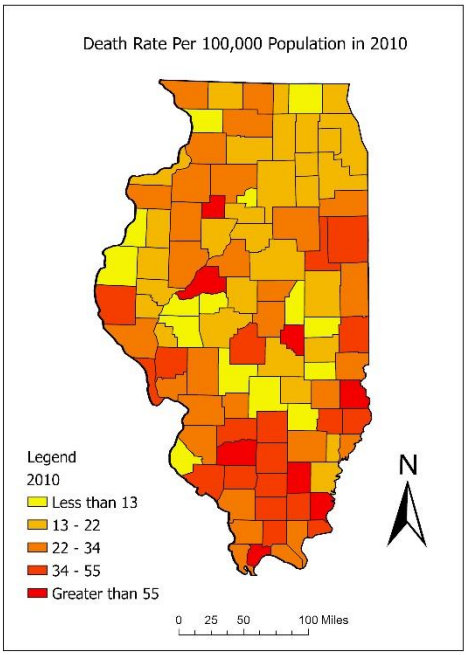
Spatial Patterns (2008-2010)

The early years exhibit a clear spatial divide across Illinois.

- Low mortality is concentrated in the northeastern region (e.g., Boone, McHenry, Lake).
- Moderate rates appear in parts of central Illinois (e.g., Peoria, McLean).
- Higher mortality is already visible in several southern counties (e.g., Pulaski, Alexander, Franklin, Perry, Gallatin, Hamilton).
  This establishes the initial spatial pattern: lower rates in the north and higher rates in the south.





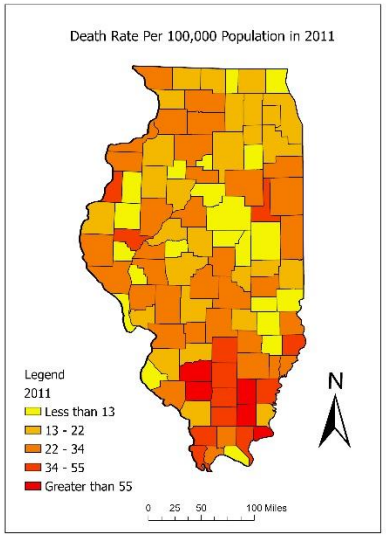


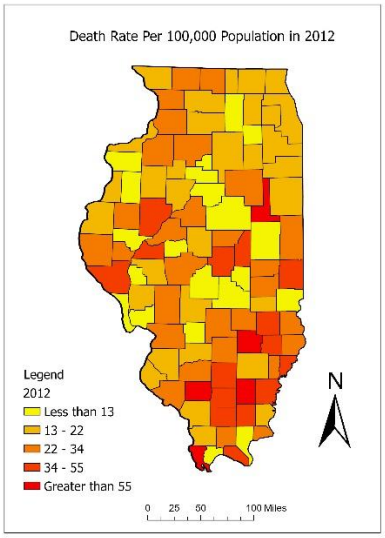
Mid-Period Trends (2011–2016)

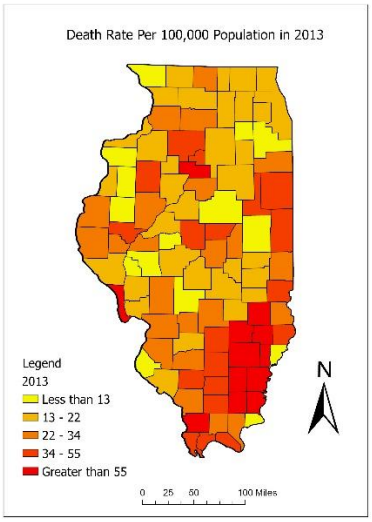
During the middle of the study period, kidney disease mortality increases in several regions.

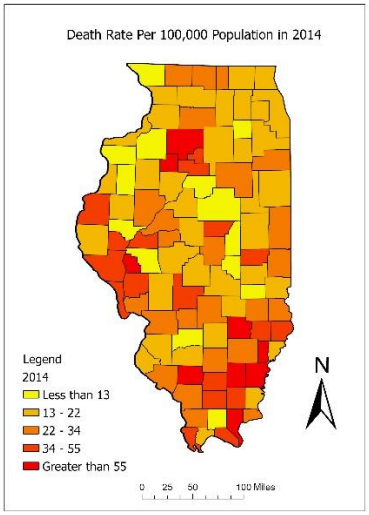
- Numerous southern and central counties shift into the moderate-to-high classifications (e.g., White, Clay, Perry, Franklin).

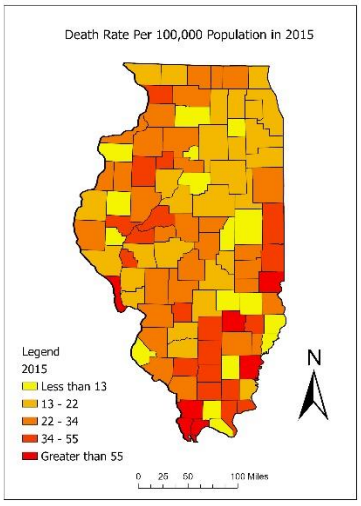
- Consistently high rates remain in the far south (e.g., Pulaski, Alexander, Saline).
- Northern counties continue to maintain lower mortality values (e.g., Putnam, Jo Daviess, Kane).
  This phase highlights the strengthening of the southern high-mortality cluster.

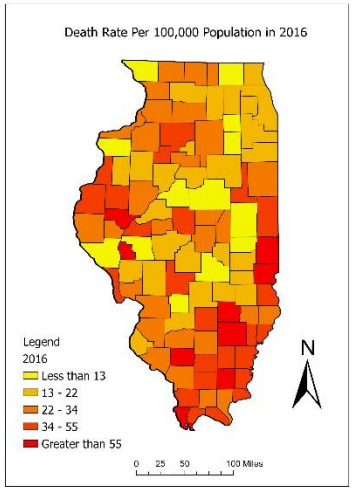










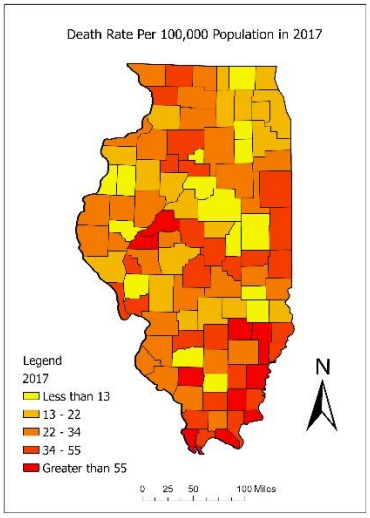


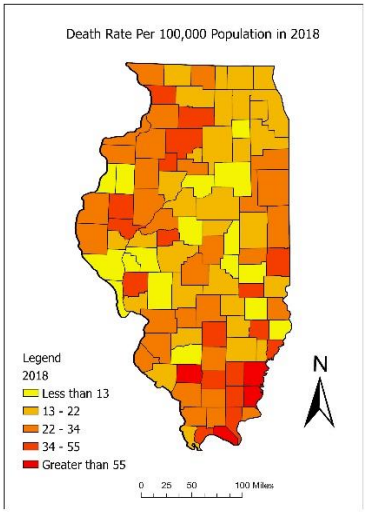
Later Trends (2017–2022)

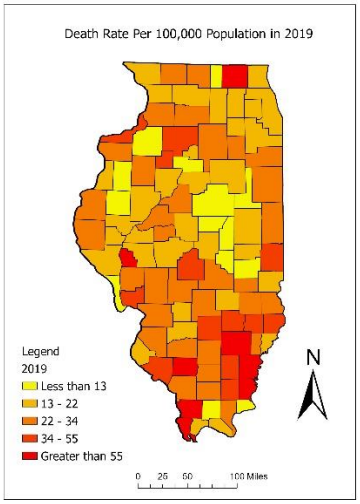
In the final years, some of the highest mortality rates appear.

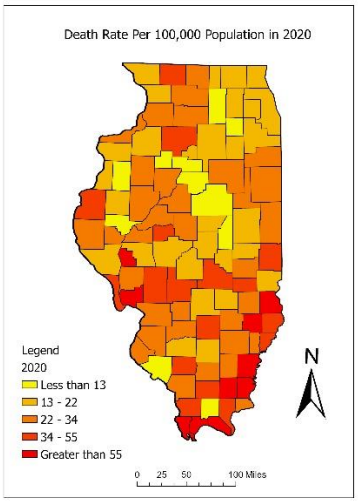
- High-mortality clusters persist in the southern region (e.g., White, Wabash).
- Several counties show peak values around 2018-2020 (e.g., Perry, Alexander, Union).

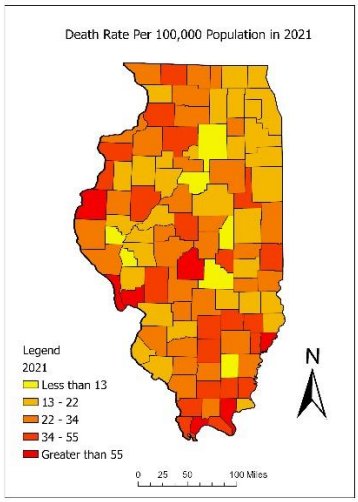
- Some counties demonstrate slight improvements in 2021-2022 (e.g., Jackson, Williamson). Despite minor fluctuations, the southern portion of the state remains the most affected throughout the entire period.

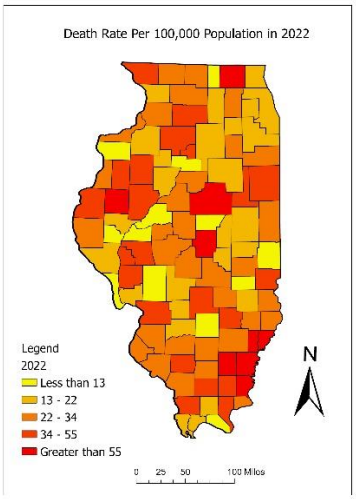






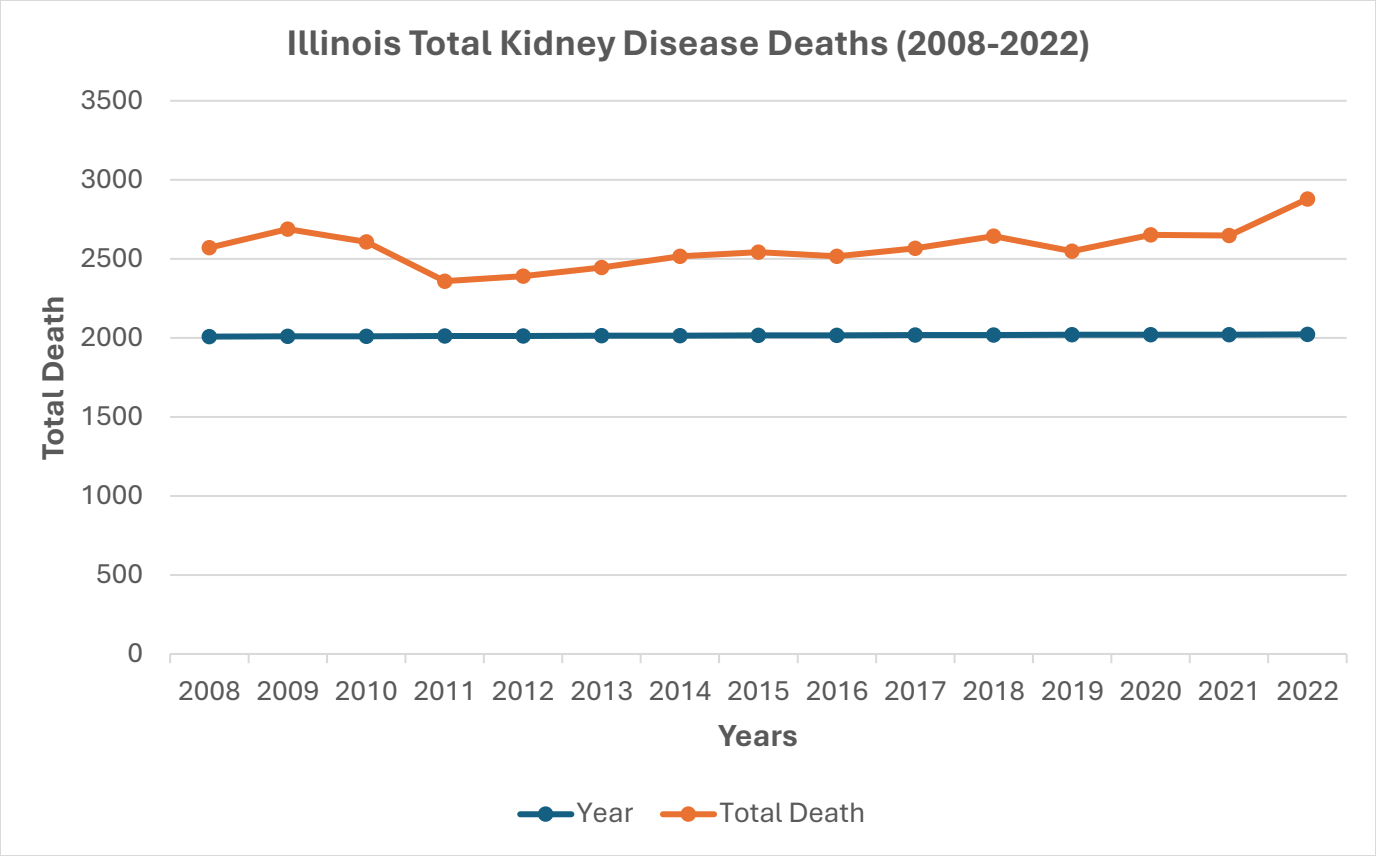






Statewide Kidney Disease Mortality Trend (2008–2022)

This graph shows the total number of kidney disease deaths per year across Illinois. Even though there are fluctuations, the general trend from 2008 to 2022 is upward. The sharp increases between 2018 and 2022 align with the hotspot growth we observed in southern Illinois counties such as White, Perry, Wabash, Pulaski, and Williamson.



Key Observations:

- Total deaths increased from 2,571 (2008) to 2,879 (2022).
- Significant dips occurred around 2011 and 2012, followed by steady increases.
- A noticeable rise happened from 2018 to 2022, matching the spatial expansion of high-mortality counties.
- 2022 recorded the highest mortality total in the entire period.

Key Findings

Across all 15 years, kidney disease mortality in Illinois demonstrates:

- Consistently elevated mortality rates in southern Illinois (e.g., Pulaski, Alexander, Johnson, Hardin, Massac).
- Persistently low mortality in northern counties (e.g., Cook, McHenry, Boone, Kendall).
- Significant spatial inequality, with a north-south gradient visible in nearly every year.
- Temporal variability, with increasing rates in the mid-decade period and mixed outcomes in later years.

The animation effectively highlights these patterns, making it easy to identify both stable clusters and shifting mortality trends. The AGOL map further supports detailed county-level exploration.

Conclusion

This project demonstrates how animated and interactive mapping techniques enhance understanding of public health trends. By standardizing classification breaks and visualizing change over time, this analysis provides meaningful insight into the geographic distribution and evolution of kidney disease mortality in Illinois.

Consistent high-rate counties may warrant further epidemiological investigation and targeted interventions. The combination of spatial analysis, time-series animation, and interactive mapping proved highly effective for communicating complex temporal disease patterns to both academic and public audiences.

Future research could incorporate socioeconomic indicators, healthcare access data, and demographic variables to explore potential drivers of spatial disparities. Expanding the visualization into an Esri StoryMap could further increase accessibility and public engagement.